

Arthur Williams CV

Number available on request | arthwills17@gmail.com | [LinkedIn](#) | [GitHub](#) | [dzladev.com](#)

PROFESSIONAL SUMMARY

Final-year Computer Science student specialising in cyber security and AI-driven threat detection. Experienced in building SOC simulations, malware classification systems, and cloud-based monitoring infrastructure. Strong foundation in Python, Linux, and security analytics.

EDUCATION

Liverpool Hope University <i>BA Computer Science (Predicted First)</i>	Liverpool 2023 – 2026
George Spencer Academy <i>A-Levels in History, Psychology, Biology; AS in Advanced Mathematics</i>	Nottingham 2021 - 2023

PROJECTS

Agentic SOC Dashboard <i>Node.js, JavaScript, Github</i>	January 2026
- Built a Node.js-based SOC simulation platform to ingest and visualise real-time security events, implementing alert prioritisation and analyst-style response workflows. - Designed monitoring logic to simulate incident triage within a full-stack dashboard environment.	
Explainable AI (XAI) Malware Detector <i>Python, TensorFlow, Jupyter, SHAP</i>	February 2026
- Trained a TensorFlow neural network on memory dump data, achieving 99% classification accuracy on test samples. - Integrated SHAP to provide feature-level explainability for analyst validation and model transparency.	
Semi-Supervised Ransomware Threat Engine <i>Python, Network Security</i>	September 2025
- Developed a semi-supervised anomaly detection model in Python achieving 99% accuracy in simulated ransomware traffic scenarios. - Analysed structured network data to generate early-warning alerts for suspicious behavioural patterns.	
Cloud-Based IoT Monitoring System <i>JavaScript, AWS, MongoDB</i>	December 2025
- Built a full-stack IoT system to monitor environmental factors like humidity and temperature in real-time. - Deployed the infrastructure on AWS, utilising API requests to fetch and store data in a MongoDB database. - Developed a web service frontend to visualise sensor data for end-users.	

TECHNICAL SKILLS

Proficient Languages: C#, Python, JavaScript
Familiar Languages: C++, C, PHP, Java, SQL (MySQL, PostgreSQL), Kotlin
Frameworks: Node.js, React.js, TensorFlow.js, HTML5/CSS3
Developer Tools: Git / GitHub, Linux (Ubuntu/Mint), AWS, VS Code, Visual Studio, IntelliJ and Android Studio.
Libraries: pandas, scikit-learn, NumPy, Matplotlib, MongoDB, MySQL, PostgreSQL

EXPERIENCE

Shift Leader at Tesco	2021 – Present
- Lead shifts in a high-volume retail environment, supervising team members and ensuring operational continuity. - Developed strong communication, incident resolution, and prioritisation skills under time pressure.	
Freelance Database Integration Specialist	2025 – Present
Supported small-scale database integration and migration tasks across SQL environments. Assisted with schema design, query optimisation, and access control configuration. Applied best practices in data integrity and performance tuning.	